# SIMATS ENGINEERING
# CSA6303- Threat Intelligence and Network Security
# A1-Malware Analysis Assignment

Assessment Questions

| | | | |
|---|---|---|---|
| **Subject** | Threat Intelligence | **Total Marks** | 30 |
| **Questions** | 3 (all compulsory) | **Marks per Question** | 10 |
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| **Faculty** | Dr.V.Parthipan | **Slot** | A |

Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

**CO3:** Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

**Code of Conduct:**

I B.Hari Sai/192311098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of Student :B.Hari Sai**

## Questions :

### Exercise 1: Threat Intelligence Platform Investigation

Objective: To investigate suspicious IP addresses, domains, and file hashes using a threat intelligence platform.

Practical Tasks:

- Examine sample IoCs using VirusTotal or another authorized threat intelligence platform.
- Review available reputation scores and threat classifications.
- Record the findings and recommend appropriate security actions.

Expected Output: A threat intelligence investigation report.

**CODE:**

```
import webbrowser

import json

while True:

print("\n1.IP Address 2.Domain 3.File Hash 4.Exit")

choice = input("Enter choice: ")

if choice == "4":

break

if choice == "1":

typ, path = "IP", "ip-address"

elif choice == "2":

typ, path = "Domain", "domain"

elif choice == "3":

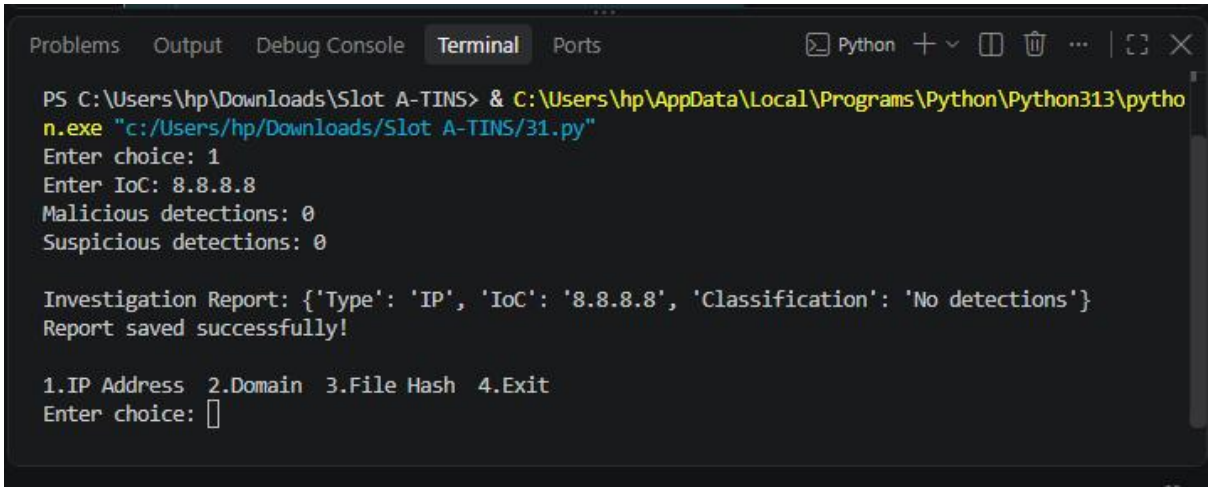
typ, path = "Hash", "file"

else:

print("Invalid choice!")

continue
```

```
ioc = input("Enter IoC: ")
webbrowser.open(f"https://www.virustotal.com/gui/{path}/{ioc}")
malicious = int(input("Malicious detections: "))
suspicious = int(input("Suspicious detections: "))
status = "Malicious" if malicious > 0 else (
"Suspicious" if suspicious > 0 else "No detections"
)
report = {
"Type": typ,
"IoC": ioc,
"Classification": status
}
with open("report.json", "a") as f:
f.write(json.dumps(report) + "\n")
print("\nInvestigation Report:", report)
print("Report saved successfully!")
```



## Exercise 2: Security Log Collection and Correlation

Objective: To collect and correlate security events from different sources.

Practical Tasks:

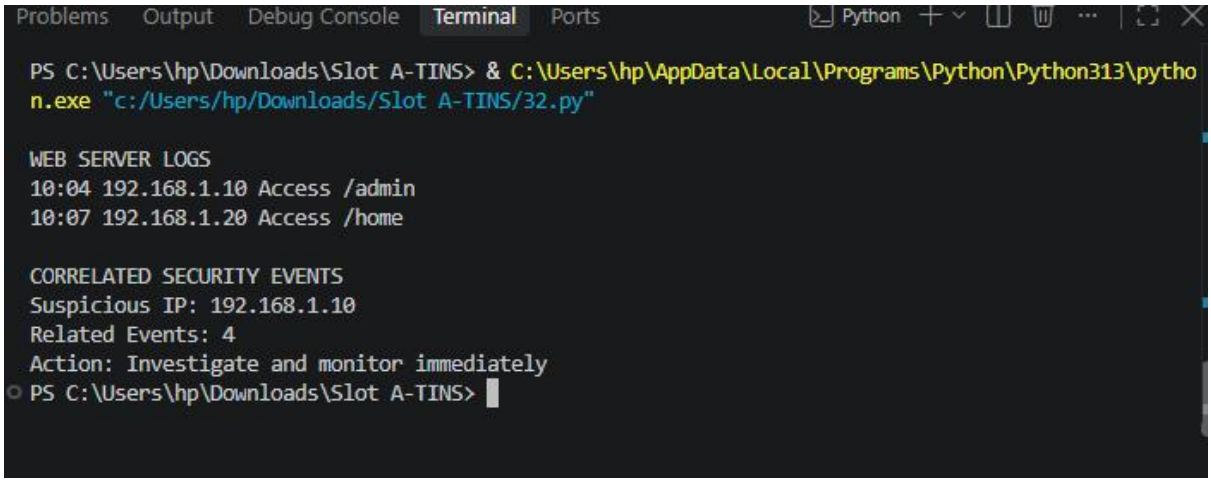
- Collect sample authentication, firewall, and web server logs.
- Identify common IP addresses, timestamps, and user activities.
- Correlate events to identify suspicious patterns.

Expected Output: A correlated security event summary.

**CODE:**

```
auth_logs = [
    "10:00 192.168.1.10 Login Failed user1",
    "10:02 192.168.1.10 Login Failed user1",
    "10:05 192.168.1.20 Login Success user2"
]
firewall_logs = [
    "10:03 192.168.1.10 Blocked Port 22",
    "10:06 192.168.1.20 Allowed Port 80"
]
web_logs = [
    "10:04 192.168.1.10 Access /admin",
    "10:07 192.168.1.20 Access /home"
]
print("AUTHENTICATION LOGS")
for log in auth_logs:
    print(log)
print("\nFIREWALL LOGS")
for log in firewall_logs:
    print(log)
print("\nWEB SERVER LOGS")
for log in web_logs:
    print(log)
all_logs = auth_logs + firewall_logs + web_logs
print("\nCORRELATED SECURITY EVENTS")
for log in all_logs:
    ip = log.split()[1]
```

```
    count = sum(ip in item.split() for item in all_logs)
    if count >= 3:
        print("Suspicious IP:", ip)
        print("Related Events:", count)
        print("Action: Investigate and monitor immediately")
        break
```




## Exercise 3: Threat Hunting Using Network Logs

Objective: To identify suspicious network activities through proactive threat hunting.

Practical Tasks:

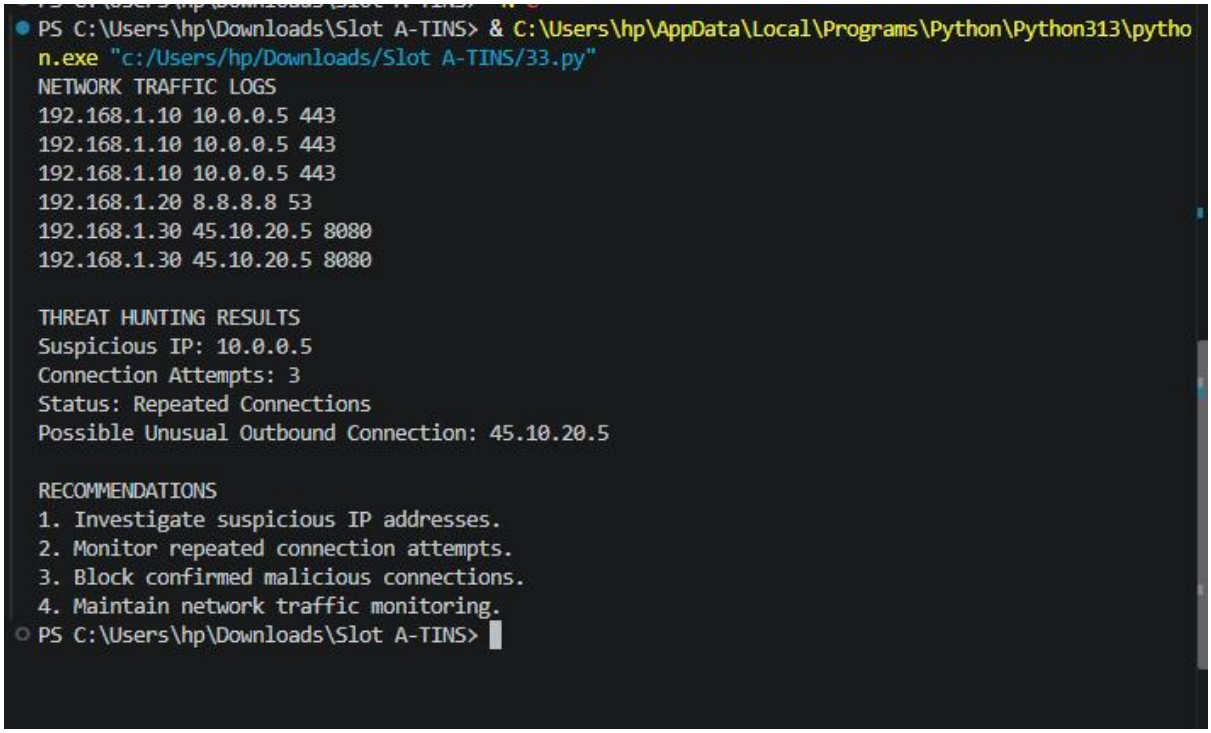
- Examine sample network traffic logs.
- Identify unusual outbound connections and repeated connection attempts.
- Investigate suspicious communication patterns and document relevant IoCs.

Expected Output: A threat-hunting report with findings and recommendations.

**CODE:**

```
logs = [
    "192.168.1.10 10.0.0.5 443",
    "192.168.1.10 10.0.0.5 443",
    "192.168.1.10 10.0.0.5 443",
    "192.168.1.20 8.8.8.8 53",
    "192.168.1.30 45.10.20.5 8080",
    "192.168.1.30 45.10.20.5 8080"
```

```
]
print("NETWORK TRAFFIC LOGS")
for log in logs:
    print(log)
print("\nTHREAT HUNTING RESULTS")
connections = {}
for log in logs:
    ip = log.split()[1]
    connections[ip] = connections.get(ip, 0) + 1
for ip, count in connections.items():
    if count >= 3:
        print("Suspicious IP:", ip)
        print("Connection Attempts:", count)
        print("Status: Repeated Connections")
    if ip == "45.10.20.5":
        print("Possible Unusual Outbound Connection:", ip)
print("\nRECOMMENDATIONS")
print("1. Investigate suspicious IP addresses.")
print("2. Monitor repeated connection attempts.")
print("3. Block confirmed malicious connections.")
print("4. Maintain network traffic monitoring.")
```


PS C:\Users\hp\Downloads\Slot A-TINS> & C:\Users\hp\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/hp/Downloads/Slot A-TINS/33.py"
NETWORK TRAFFIC LOGS
192.168.1.10 10.0.0.5 443
192.168.1.10 10.0.0.5 443
192.168.1.10 10.0.0.5 443
192.168.1.20 8.8.8.8 53
192.168.1.30 45.10.20.5 8080
192.168.1.30 45.10.20.5 8080
THREAT HUNTING RESULTS
Suspicious IP: 10.0.0.5
Connection Attempts: 3
Status: Repeated Connections
Possible Unusual Outbound Connection: 45.10.20.5
RECOMMENDATIONS
1. Investigate suspicious IP addresses.
2. Monitor repeated connection attempts.
3. Block confirmed malicious connections.
4. Maintain network traffic monitoring.
PS C:\Users\hp\Downloads\Slot A-TINS>